

Laboratory Report 07

Course: B.Sc. (Hons) in IT - Database Systems

Topic: Using Oracle's EXPLAIN PLAN

1. Introduction

The main objective of this laboratory report is to study how to use EXPLAIN PLAN to observe the execution plans chosen by the Oracle optimizer for SELECT, UPDATE, INSERT, and DELETE statements.

2. Initial Setup

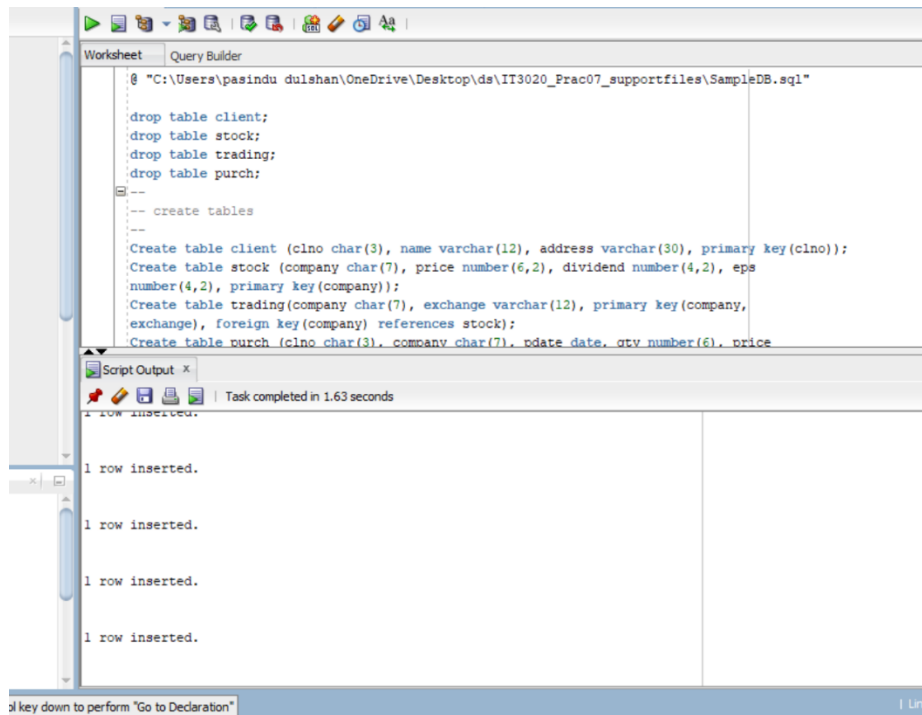
First, the database and required tables need to be set up.

1. The SampleDB.sql script was executed to create the client, stock, trading, and purch tables and insert sample data into them.
2. The utlxplan.sql script was executed to create the PLAN_TABLE, which is essential for storing the query plan.
3. The session was configured to view CPU and I/O costs using the following commands:

SQL

```
ALTER SESSION SET OPTIMIZER_MODE = ALL_ROWS;
```

```
ALTER SESSION SET "_optimizer_cost_model" = CPU;
```



3. Part (A) & (B): Checking Query Plan Without Indexes

The query plan was examined for retrieving the client number and name of clients who purchased more than 1000 shares, without any additional indexes.

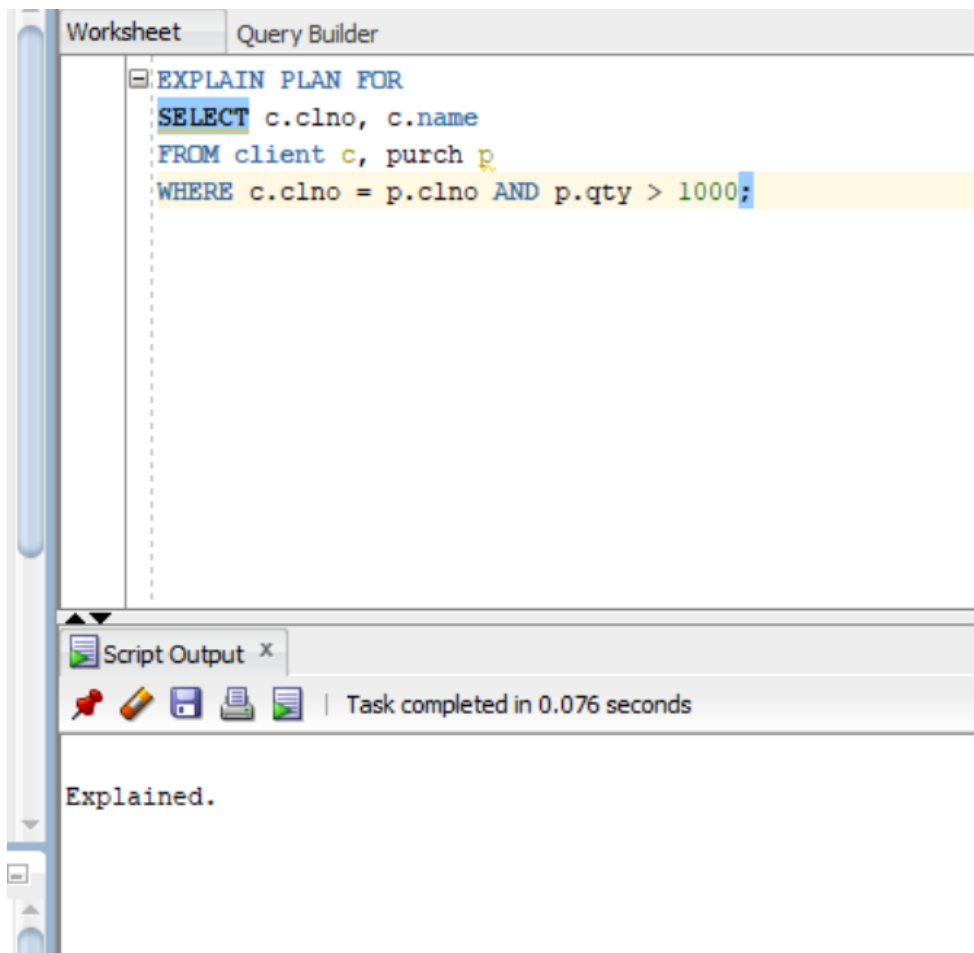
SQL

EXPLAIN PLAN FOR

SELECT c.cno, c.name

FROM client c, purch p

WHERE c.cno = p.cno AND p.qty > 1000;



Then, the utlxpls.sql script was executed to view the stored plan and its costs.

The screenshot shows the Oracle SQL Developer interface. The top toolbar includes icons for running queries, saving, and other standard database operations. Below the toolbar, the 'Worksheet' tab is active, displaying the following SQL query:

```
WHERE c.clno = p.clno AND p.qty > 1000;
```

The 'Query Builder' tab is also visible. Below the query, the 'Script Output' and 'Query Result' tabs are shown. The 'Query Result' tab displays the execution plan for the query, which is a 'Plan Table'.

Operation and options	Object	cost	cpu_cost	io_cost
SELECT STATEMENT		3	16313	3
HASH JOIN		3	16313	3
NESTED LOOPS		3	16313	3
NESTED LOOPS		3	16313	3
STATISTICS COLLECTOR				
TABLE ACCESS FULL	PURCH	2	17121	2
INDEX UNIQUE SCAN	SYS_C0083	0	1900	0
TABLE ACCESS BY INDEX ROWID	CLIENT	1	9191	1
TABLE ACCESS FULL	CLIENT	1	9191	1

Below the plan table, the text '12 rows selected.' is displayed.

Query Plan Description:

- **INDEX UNIQUE SCAN:** First, Oracle uses the primary key index of the Client table to find the clno. The cost for this is 0.
- **TABLE ACCESS BY INDEX ROWID:** Using the ROWID found from the index, it retrieves the name from the Client table. The cost is 1.
- **TABLE ACCESS FULL on PURCH:** Oracle reads the entire PURCH table (row by row) to check if qty > 1000. Since it reads the whole table, the cost increases to 2.
- **NESTED LOOPS:** Steps 1 and 2 are joined for each row found in the PURCH table to find the relevant client's name.

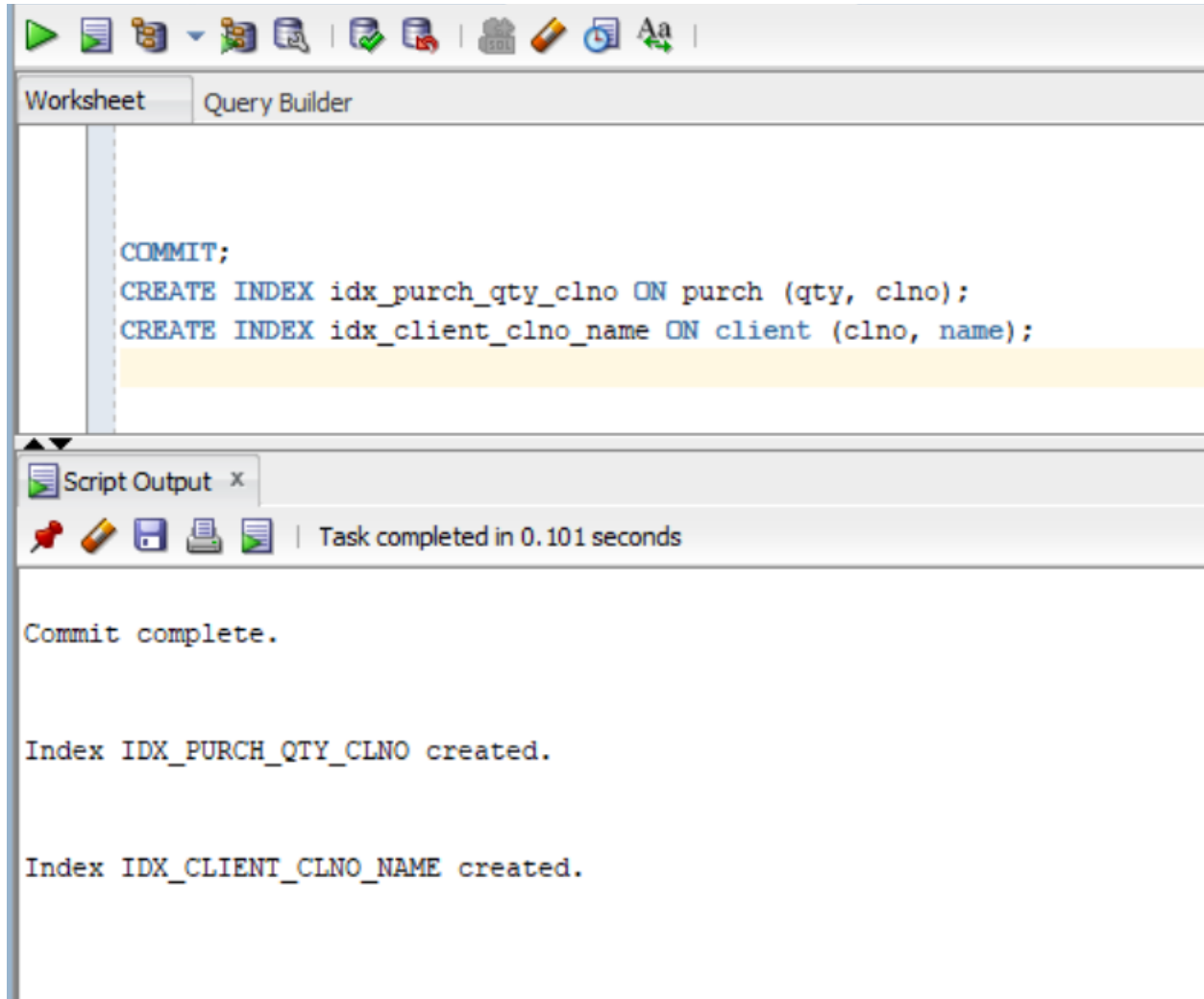
4. Part (C): Creating Indexes

To speed up the data retrieval process, Unclustered B+-Tree Indexes were created for the PURCH and CLIENT tables.

SQL

```
CREATE INDEX idx_purch_qty_clno ON purch (qty, clno);
```

```
CREATE INDEX idx_client_clno_name ON client (clno, name);
```



5. Part (D): Comparing Query Plan With Indexes

With the newly created indexes, the EXPLAIN PLAN command from Part (A) was executed again, and the new query plan was obtained via utlxpls.sql.

Worksheet Query Builder

```
CREATE INDEX idx_client_clno_name ON client (clno, name);
```

EXPLAIN PLAN FOR

```
SELECT c.clno, c.name
FROM client c, purch p
WHERE c.clno = p.clno AND p.qty > 1000;
```

Script Output x

Task completed in 0.07 seconds

Explained.

Plan Table

Operation and options	Object	cost	cpu_cost	io_cost
SELECT STATEMENT		0	1250	0
NESTED LOOPS		0	1250	0
NESTED LOOPS		0	1250	0
INDEX RANGE SCAN	IDX_PURCH	0	200	0
INDEX UNIQUE SCAN	SYS_C0083	0	1050	0
TABLE ACCESS BY INDEX ROWID	CLIENT	0	1050	0

9 rows selected.

Explained.

Comparison and Conclusion:

- **Purch Table Access:**
 - *Without Index (Part B):* **TABLE ACCESS FULL:** Reads all 10 rows in the entire table.
 - *With Index (Part D):* **INDEX RANGE SCAN:** The index quickly finds only the 3 rows where qty > 1000.
- **Client Table Access:**
 - *Without Index (Part B):* **TABLE ACCESS BY INDEX ROWID:** Must go to the table to retrieve the name.
 - *With Index (Part D):* **Covering Index:** The name is already included in the index, so there is no need to visit the separate table.